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ASSIGNMENT-02

1] prove that the spherical coordinate system is Orthogonal?

→ we have for the spherical system.

$\vec{r} = r\sin\theta\cos\phi\, i + r\sin\theta\sin\phi\, j + r\cos\theta\, k$

$\hat{e}_r, \hat{e}_\theta, \hat{e}_\phi$ these are unit of spherical system.

$\hat{e}_r = \frac{1}{h_1}\left|\frac{\partial \vec{r}}{\partial r}\right| \quad \hat{e}_\theta = \frac{1}{h_2}\left|\frac{\partial \vec{r}}{\partial \theta}\right| \quad \hat{e}_\phi = \frac{1}{h_3}\left|\frac{\partial \vec{r}}{\partial \phi}\right|$

$h_1 = 1 \qquad h_2 = r \qquad h_3 = r\sin\theta$

$\hat{e}_r = \sin\theta\cos\phi\, i + \sin\theta\sin\phi\, j + \cos\theta\, k$

$\hat{e}_\theta = r\cos\theta\cos\phi\, i + r\cos\theta\sin\phi\, j + (-r\sin\theta)\hat{k}$

$\hat{e}_\phi = -r\sin\theta\sin\phi\, i + r\sin\theta\cos\phi + 0k.$

$\hat{e}_r = \frac{1}{1}\sin\theta\cos\phi\, i + \sin\theta\sin\phi\, j + \cos\theta\, k \rightarrow (1)$

$\hat{e}_\theta = \frac{1}{r}\left(r(\cos\theta\cos\phi\, i + \cos\theta\sin\phi\, j + (-\sin\theta)k\right)$

$= \cos\theta\cos\phi\, i + \cos\theta\sin\phi\, j - \sin\theta k. \rightarrow (2).$

$\hat{e}_\phi = \frac{1}{r\sin\theta}\left(-r\sin\theta\sin\phi\, i + r\sin\theta\cos\phi + 0k\right)$

$= \frac{r\sin\theta}{r\sin\theta}\left(-\sin\phi + \cos\phi + 0k\right)$

$= -\sin\phi + \cos\phi + 0k // \rightarrow (3).$

$\hat{e}_r \cdot \hat{e}_\theta = \sin\theta\cos\theta\cos^2\theta + \sin\theta\cos\theta\sin^2\phi - \sin\theta\cos\theta$

$= \sin\theta\cos\theta(\cos^2\phi + \sin^2\phi) - \sin\theta\cos\theta$

$= 0.$

$\hat{e}_\theta \cdot \hat{e}_\phi = -\cos\theta\cos\phi\sin\phi + \cos\theta\sin\phi\cos\phi + 0$

$= 0$

$\hat{e}_\phi \hat{e}_r = -\sin\theta\cos\phi\sin\phi + \sin\theta\sin\phi\cos\phi + 0 = 0.$

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2] prove that the cylindrical coordinate system is Orthogonal?

=> we have in cylindrical system.

$x = r\cos\theta \quad y = r\sin\theta \quad z = z$

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$\vec{r} = (r\cos\theta)\hat{i} + (r\sin\theta)\hat{j} + (z)\hat{k}$

$\hat{e_r} = \dfrac{\frac{\partial \vec{r}}{\partial r}}{\left|\frac{\partial \vec{r}}{\partial r}\right|} = \dfrac{\cos\theta\hat{i} + \sin\theta\hat{j} + 0\hat{k}}{\sqrt{\cos^2\theta + \sin^2\theta + 0}}$

$= \cos\theta\hat{i} + \sin\theta\hat{j} + 0\hat{k}$ //

$\hat{e_\theta} = \dfrac{\frac{\partial \vec{r}}{\partial \theta}}{\left|\frac{\partial \vec{r}}{\partial \theta}\right|} = \dfrac{-r\sin\theta\hat{i} + r\cos\theta\hat{j} + 0\hat{k}}{\sqrt{r^2\sin^2\theta + r^2\cos^2\theta + 0}}$

$= \dfrac{r(-\sin\theta\hat{i} + \cos\theta\hat{j})}{\sqrt{r^2}}$

$= -\sin\theta\hat{i} + \cos\theta\hat{j}$ //

$\hat{e_z} = \dfrac{\frac{\partial \vec{r}}{\partial z}}{\left|\frac{\partial \vec{r}}{\partial z}\right|} = \hat{k}$ //

$\hat{e_r}\,\hat{e_\theta} = (\cos\theta\hat{i} + \sin\theta\hat{j} + 0\hat{k}) \cdot (-\sin\theta\hat{i} + \cos\theta\hat{j}) + 0\hat{k}$

$= -\sin\theta \cdot \cos\theta + \sin\theta \cdot \cos\theta + 0$

$= 0$ //

$\hat{e_\theta}\,\hat{e_z} = (-\sin\theta\hat{i} + \cos\theta\hat{j} + 0\hat{k})\,(0\hat{i} + 0\hat{j} + 0\hat{k})$

$= 0$ //

$\hat{e_z} \cdot \hat{e_r} = (0\hat{i} + 0\hat{j} + 0\hat{k})\,(\cos\theta\hat{i} + \sin\theta\hat{j} + 0\hat{k})$

$= 0$ //

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3] Express the vector $\vec{A} = zi + 2xj + yk$ in cylindrical coordinates ?

⇒ Let $\vec{A} = A_1\hat{e}_\rho + A_2\hat{e}_\phi + A_3\hat{e}_z$ → (1).

$A_1 = \vec{A}\vec{e}_\rho$ $A_2 = \vec{A}\vec{A}_\phi$ $A_3 = \vec{A}\vec{e}_z$

for cylindrical co-ordinates.

$x = \rho\cos\phi \quad y = \rho\sin\phi \quad z = z$ → (2).

$\vec{r} = \rho\cos\phi\, i + \rho\sin\phi\, j + zk$

$\hat{e}_\rho = \frac{1}{h_1}\frac{\partial\vec{r}}{\partial\rho}$ $\hat{e}_\phi = \frac{1}{h_2}\frac{\partial\vec{r}}{\partial\phi}$ $\hat{e}_z = \frac{1}{h_3}\frac{\partial\vec{r}}{\partial z}$

$h_1 = 1$ $h_2 = \rho$ $h_3 = 1$

$\hat{e}_\rho = \cos\phi\, i + \sin\phi\, j$

$\hat{e}_\phi = \frac{1}{\rho}(-\rho\sin\phi\, i + \rho\cos\phi\, j + 0k)$

$= -\sin\phi\, i + \cos\phi\, j.$

$\hat{e}_z = 1k.$

use eq (2) values in eq (1).

$\vec{A} = zi + 2\rho\cos\phi\, j + \rho\sin\phi k$

$A_1 = \vec{A}\hat{e}_\rho = z\cos\phi - 2\rho\sin\phi\cos\phi$

$A_2 = \vec{A}\hat{e}_\phi = -z\sin\phi - 2\rho\cos^2\phi$

$A_3 = \vec{A}\hat{e}_z = \rho\sin\phi.$

$\vec{A} = (z\cos\phi - 2\rho\sin\phi\cos\phi)\hat{e}_\rho - (z\sin\phi + 2\rho\cos^2\phi)e_\phi + \rho\sin\phi\cdot\hat{e}_z.$

4] Determine whether the matrix $\begin{bmatrix} -1 & 7 \\ 8 & -1 \end{bmatrix}$ is a

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Linear combination of $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$ in the vector space $M_{22}$ of $2 \times 2$ matrices?

$\Rightarrow$ $V = c_1 A + c_2 B + c_3 C$.

$$\begin{bmatrix} -1 & 7 \\ 8 & -1 \end{bmatrix} = c_1 \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 & -3 \\ 0 & 2 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 7 \\ 8 & -1 \end{bmatrix} = \begin{bmatrix} c_1 + 2c_2 & -3c_2 + c_3 \\ 2c_1 + 2c_3 & c_1 + 2c_2 \end{bmatrix}$$

$c_1 + 2c_2 = -1 \rightarrow (1)$

$-3c_2 + c_3 = 7 \rightarrow (2)$

$2c_1 + 2c_3 = 8 \rightarrow (3)$

$c_1 + 2c_2 = -1 \rightarrow (4)$.

eq (3) $-$ 2(eq (2))

$2c_1 + 2c_3 = 8$

$-6c_2 + 2c_3 = 14$

$2c_1 + 6c_2 = -6$.

(4) $-$ 2 (1).

$2c_1 + 6c_2 = -6$

$2c_1 + 4c_2 = -2$

$2c_2 = -4$

$\boxed{c_2 = -2}$

$c_1 + 2(-2) = -1$

$c_1 = -1 + 4$

$\boxed{c_1 = 3}$

$-3(-2) + c_3 = 7$

$6 + c_3 = 7$

$\boxed{c_3 = 1}$

5] Decide whether or not the vector $M = \begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix}$ in the vector space of second order square matrices is a linear combination of the following matrices.

$M_1 = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$ $M_2 = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$ and $M_3 = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$

$\Rightarrow$ $A = c_1 B + c_2 C + c_3 D$.

$$\begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix} = c_1 \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} + c_2 \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

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$$\begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} c_1+c_2+c_3 & c_1+c_2-c_3 \\ -c_2 & -c_1 \end{bmatrix}$$

$c_1 + c_2 + c_3 = 3$

$c_1 + c_2 - c_3 = -1$

$-c_2 = 1$

$-c_1 = -2$

$\boxed{c_2 = -1}$ $\boxed{c_1 = 2}$

Thus A is a Linear combination.

$2 + (-1) - c_3 = -1$

$2 - 1 - c_3 = -1$

$1 - c_3 = -1$

$-c_3 = -1 - 1$

$\boxed{c_3 = 2}$

$A = 2B - 1C + 2D$ //

6] Is the set of vectors $\{(1,2,4)(1,0,0)(0,1,0)(0,0,1)\}$ linearly independent?

=> $A \sim \begin{bmatrix} 1 & 2 & 4 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $R_2 \to R_2 - R_1$

$A \sim \begin{bmatrix} 1 & 2 & 4 \\ 0 & -2 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $R_3 \to 2R_3 + R_2$

$A \sim \begin{bmatrix} 1 & 2 & 4 \\ 0 & -2 & -4 \\ 0 & 0 & -4 \\ 0 & 0 & 1 \end{bmatrix}$ $R_4 \to 4R_4 + R_3$

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$$A \sim \begin{bmatrix} 1 & 2 & 4 \\ 0 & -2 & -4 \\ 0 & 0 & -4 \\ 0 & 0 & 0 \end{bmatrix} \quad R_4 \to \text{zero row}$$

The given set of vector is not linear independent therefore it is dependent.

7] Determine whether the vector $V_1 = (1,2,3)$ $V_2 = (3,1,7)$ $V_3 = (2,5,8)$ are linearly dependent or linearly independent.

⇒ $$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 7 \\ 2 & 5 & 8 \end{bmatrix}$$

$|A| = [1(8-35) - 2(24-14) + 3(15-2)]$

$= [1(-27) - 2(10) + 3(13)$

$= |-27 - 20 + 39|$

$= -8$

$|A| \neq 0$.

Therefore the given vectors are linearly independent.

8]. find the basis and the dimension of the subspace spanned by the vectors

$\{(2,4,2), (1,-1,0), (1,2,1), (0,3,1)\}$ in $V_3(R)$

⇒ $$A \sim \begin{bmatrix} 2 & 4 & 2 \\ 1 & -1 & 0 \\ 1 & 2 & 1 \\ 0 & 3 & 1 \end{bmatrix} \quad R_1 \leftrightarrow R_2.$$

$$A \sim \begin{bmatrix} 1 & -1 & 0 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \\ 0 & 3 & 1 \end{bmatrix} \quad \begin{matrix} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - R_1 \end{matrix}$$

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$$A \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 3 & 2 \\ 0 & 3 & 1 \\ 0 & 3 & 1 \end{bmatrix} \quad R_2 \leftrightarrow R_3$$

$$A \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 3 & 1 \\ 0 & 3 & 2 \\ 0 & 3 & 1 \end{bmatrix} \quad \begin{matrix} R_3 \to R_3 - 2(R_2) \\ R_4 \to R_4 - R_3 \end{matrix}$$

$$A \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

subspace = $\{(1,-1,0)\ (0,3,1)\ (0,0,-1)\}$

dimension = 1

9] find the basis and the dimension of the subspace spanned by the $\{(0,1,-3,-1)\ (1,0,1,1)\ (3,1,0,2)\ (1,1,-2,0)\}$ Vectors?

$$\Rightarrow A \sim \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix} \quad R_2 \leftrightarrow R_1$$

$$A \sim \begin{bmatrix} 1 & 1 & -2 & 0 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 0 & 1 & -3 & -1 \end{bmatrix} \quad \begin{matrix} R_3 \to R_3 - 3R_1 \\ R_2 \to R_2 - R_1 \end{matrix}$$

$$A \sim \begin{bmatrix} 1 & 1 & -2 & 0 \\ 0 & -1 & 3 & 1 \\ 0 & -2 & 6 & 2 \\ 0 & 1 & -3 & -1 \end{bmatrix} \quad \begin{matrix} R_3 \to R_3 - 2R_2 \\ R_4 \to R_4 + R_2 \end{matrix}$$

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$$A \sim \begin{bmatrix} 1 & 1 & -2 & 0 \\ 0 & -1 & 3 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Subspace = $\{(1,1,-2,0), (0,-1,3,1)\}$

dimension = 2.

10] Find the matrix of the linear Transformation $T: V_2(R) \to V_3(R)$ such that $T(-1,1) = (-1,0,2)$ $e_1$ $e_2$ 11]

$T(2,1) = (1,2,1)$

$\Rightarrow$ Let $(-1,1) = (-1,0,2) = -1e_1 + e_2 = -e_1 + e_2$

$\therefore (2,1) = 2e_1 + e_2$

Transformation of $(-1,1) = T(-e_1 + e_2)$

$= -T(e_1) + T(e_2)$

$T(-1,1) = -T(e_1) + T(e_2)$

$(-1,0,2) = -T(e_1) + T(e_2) \to (1)$.

$T(2,1) = T(2e_1 + e_2) = 2T(e_1) + T(e_2)$

$T(2,1) = 2T(e_1) + T(e_2)$

$(1,2,1) = 2T(e_1) + T(e_2) \to (2)$.

eq (1) - eq (2).

$(-2,-2,1) = -3T(e_1)$

$(2,2,-1) = 3T(e_1)$ //.

11] ~~prove that the transformation $T: R2 \to R2$ defined by $T(x,y) = (3x, x+y)$ is linear. Find the images of the vectors $(1,3)$ and $(-1,2)$ under this transformation.~~

$T(e_1) = \left(\frac{2}{3}, \frac{2}{3}, \frac{-1}{3}\right)$ //.

Substitute in eq (2).

$(1,2,1) = 2T(e_1) + T(e_2)$

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$T(e_2) = \left(\frac{2}{3}, \frac{2}{3}, \frac{-1}{3}\right) + (-1, 0, 2)$

$T(e_2) = \left[\frac{-1}{3}, \frac{2}{3}, \frac{5}{3}\right]$

Matrix of $L_0T = \begin{bmatrix} 2/3 & -1/3 \\ 2/3 & 2/3 \\ -1/3 & 5/3 \end{bmatrix}$ //

11] Prove that the transformation $T: R2 \to R2$ defined by $T(x,y) = (3x, x+y)$ is linear. Find the images of the vectors $(1,3)$ and $(-1,2)$ under this transformation?

=> $T(1,3) = (3,4,0)$

$T(-1,2) = (-3,1,0)$

let $(1,3) = (3,4,0) = 3e_1 + 4e_2$

$(-1,2) = (-3,1,0) = -3e_1 + 4e_2$

Transformation of $(3,4) = T(3e_1 + 4e_2)$

$T(3,4,0) = 3T(e_1) + 4T(e_2) \to (1)$.

$T(-3,1,0) = -3T(e_1) + T(e_2) \to (2)$.

eq (1) - eq (2).

$(0,3,0) = 3T(e_2)$

$(0,3,0) = 3T(e_2)$

$T(e_2) = (0,1,0)$ //

substitute in eq (2)

$(3,1,0) = -3T(e_1) + T(e_2)$

$-3Te_1 = (3,1,0) - (0,1,0)$

$-3Te_1 = (3,0,0)$

$T(e_1) = (-1,0,0)$

$L_0T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}$ //

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12] If $S: V_3(R) \to V_2(R)$ defined by $S(x,y) = (3x+2y, 3x-4y)$ show that S is linear but T is not linear.

⇒ $S: V_3(R) \to V_2(R)$

$a = (a_1, b_1)$ $b = (a_2, b_2)$

$T(a) = T(a_1, b_1) = (3a_1+2b_1, 3a_1-4b_1)$

$T(b) = (3a_2+2b_2, 3a_2-4b_2)$

Consider.

$T(a+b) = T(a_1+a_2, b_1+b_2)$

$= T(3a_1+3a_2+2b_1+2b_2, 3a_1+3a_2-4b_1-4b_2)$

$= T[3a_1+2b_1+3a_2+2b_2, 3a_1+4b_1+3a_2-4b_2)$

$= T[3a_1+2b_1, 3a_1-4b_1) + (3a_2+2b_2, 3a_2-4b_2)$

$= S(a) + S(b)$.

$S(c.a) = S(ca_1, cb_1)$

$= 3ca_1+2cb_1, 3ca_1-4cb_1$

$= c[3a_1+2b_1), c(3a_1-4b_1)$

$= c[3a_1+2b_1, 3a_1-4b_1)$

$= c\,S(a)$ //

13]. find the kernel and range of the linear Operator?

⇒ $T(x,y,z) = (x+y, z)$ of $R_3 \to R_2$

(i) kernel => $T(x,y,z) = (0,0)$

$(x+y, z) = (0,0)$

$x+y=0$, $z=0$

$y = -x$ $z=0$

kernel $(T) = (x, -x, 0)$

kernel $(T) = \{(1,-1,0)\}$

(ii) $T(x,y,z) = (x+y, z)$

let $a = x+y$ and $b = z$

range $(T) = \{(a,b) | a,b \in R\} = R^2$

range(T) = $R^2$

14] Verify the Rank - nullity theorem for the linear transformation?

Q T: R3 → R3 defined by T(x, y, z) = (x+y, x-y, 2x+z)

=> Let T: V → W

T(x, y, z) = (x+y, x-y, 2x+z)

let $e_1 = (1,0,0)$ $e_2 = (0,1,0)$ $e_3 = (0,0,1)$

$T(e_1) = T(1,0,0) = (1,1,2)$

$T(e_2) = T(0,1,0) = (1,-1,0)$

$T(e_3) = T(0,0,1) = (0,0,1)$

we should consider the transpose of the matrix of L.T

consider $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $R_2 \to R_2 - R_1$

$A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & -2 \\ 2 & 0 & 1 \end{bmatrix}$

R(T) is the subspace generated by (1,1,2) (0,-2,-2) & (0,0,1)

∴ Rank = 3 ∴ nullity = 0.

Rank + Nullity = dim V

3 + 0 = 3

3 = 3 //.

Rank Nullity theorem is verified.

Range $= x_1(1,1,2) + x_2(0,-2,-2) + x_3(0,0,1)$

$R(T) = (x_1, x_1 - 2x_2, 2x_1 - 2x_2 + x_3)$ //.

To find N(T) consider.

$T(x, y, z) = (0,0,0)$

$(x+y, x-y, 2x+z) = (0,0,0)$

$x+y=0$, $x-y=0$, $2x+z=0$.

$x=y$, $x=0$, $y=0$, $z=0$ //.

Therefore $n(T) = \{(0,0,0)\}$.

15] Verify the Rank-nullity theorem for the linear tranformation
$T: R3 \to R3$ defined by $T(x,y,z) = (x+2y-z, y+z, x+y-2z)$.

⇒ $T(x,y,z)$

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 1 & 1 & -2 \end{bmatrix} \quad R_3 \to R_3 - R_1$$

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix} \quad R_3 + R_2.$$

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

rank(T) = 2. nullity = 1

Rank + nullity = dimV.

2 + 1 = 3

3 = 3 //.

Rank Nullity theorem is verified.

Range = $x_1(1,2,-1) + x_2(0,1,1) + x_3(0,0,0)$.

$R(T) = (x_1, 2x_1+x_2, -x_1+x_2)$.

To find n(T) conider.

$T(x,y,z) = (0,0,0)$

$(x+2y-z, y+z, x+y-2z) = (0,0,0)$

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$x+2y-z=0 \quad y+z=0 \quad x+y-2z=0$

$\boxed{y=-z} \quad x-z-2z=0 \quad \boxed{z=z}$

$x-3z=0$

$\boxed{x=3z}$

$N(T) = T(x,y,z) = \{3z, -z, z\}$ //.

16] Let $T: V_3(R) \to V_2(R)$ definide by $T(x,y,z) = (y-x, y-z)$. Verify the Rank-nullity theorem for the linear transformation.

=> Let $T: V_3(R) \to V_2(R)$.

Let $e_1 = (1,0,0) \quad e_2 = (0,1,0) \quad e_3 (0,0,1)$

$T(e_1) = T(1,0,0) = (-1,0)$

$T(e_2) = T(0,1,0) = (1,1)$

$T(e_3) = T(0,0,1) = (0,-1)$

$$A = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 0 & -1 \end{bmatrix} \quad R_2 \to R_2 + R_1$$

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \\ 0 & -1 \end{bmatrix} \quad R_3 \to R_3 + R_2$$

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$R(T) = \{x_1(-1,0) + x_2(0,1)\}$

$R(T) = \{(-x_1, x_2)\}$ is the Range space.

$\dim R(T) = 2$

$\rho(T) = 2$

To find $N(T)$

consider

$T(x,y,z) = (0,0) \Rightarrow (y-x, y-z) = (0,0)$

$y-x=0$ $\quad y-z=0$

$y=x$ $\quad y=z$

$x=y=z=k.$

$T(x,y,z) = \{(k,k,k)\} \quad k \in R$ //

$\dim N(T) = 1$

$n(T) = 1$

$\dim V = \text{Rank} + \text{nullity}$

$3 = 2+1$

$3 = 3$ //.

17] Show that the function $f(x) = 3x-2$ $g(x) = x$ are orthogonal in $P_n$ with inner product of $\langle f,g \rangle = \int_0^1 f(x)\,g(x)\,dx.$

$\Rightarrow \langle f,g \rangle = \int_0^1 (3x-2)\,x\,dx$

$= \int_0^1 (3x^2-2x)\,dx.$

$= \left[ \frac{3x^3}{3} - \frac{2x^2}{2} \right]_0^1 dx$

$= x^3 - x^2 \Big]_0^1$

$= 1-1\,(0-0)$

$= 0$ //.

18] Show that the vectors $a=(2,1,3)$ $b=(-6/7, 15/14, 3/14)$ $c=(1/3, 1/3, -1/3)$ are mutually orthogonal?

$\Rightarrow$ Inner product of $\langle a,b \rangle = 2\left(\frac{-6}{7}\right) + 1\left(\frac{15}{14}\right) + 3\left(\frac{3}{14}\right)$

$= \frac{-12}{7} + \frac{15}{14} + \frac{9}{14} = 0.$

Inner product of $\langle b,c \rangle = \frac{-6}{7}\left(\frac{1}{3}\right) + \frac{15}{14}\left(\frac{1}{3}\right) + \frac{3}{14}\left(\frac{-1}{3}\right)$

$$= \frac{-6}{21} + \frac{15}{42} - \frac{3}{42} = 0.$$

Inner product of $\langle c|a \rangle = \frac{1}{3}(2) + \frac{1}{3}(1) - \frac{1}{3}(3)$

$$= \frac{2}{3} + \frac{1}{3} - 1 = 0.$$

Since $\langle a|b \rangle = \langle b|c \rangle = \langle c|a \rangle = 0$ //.